Fallacies in Argumentation on the Web

Annotating Fallacies in User Generated Content

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Definition of Fallacy

"Hitler was a vegetarian, therefore, I don't trust vegetarians."

"If we allow gay people to get married, what's next? Allowing people to marry their dogs?"

"All cats die; Socrates died; therefore Socrates was a cat."

- Fallacies generally refer to the logical or linguistic flaws in speech which states a claim along its reasons, in an attempt to form an argument.

- Fallacies are common in everyday life and thus in user generated texts as well.

- They can be regarded noise output from the generation of ideas, realisation of communication or production of knowledge.

The Need to Detect Fallacies

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- Improves argumentation detection systems
- Can be embedded in real-world applications such as search engines, question answering systems, writing assistance systems

- An emerging field in NLP and there are no studies except a few position papers:
    - argument markup language [1],
    - a formal language inspired from automated theorem proving [2]
- There is a good deal of qualitative analysis of fallacies [3, 4, 5]

[1] Gibson, A., Rowe, G. & Reed, C. (2007). A Computational Approach to Identifying Formal Fallacy. Workshop on Computational Models of Natural Argument.
[2] Reed, C., & Koszowy, M. (2011). The development of argument and computation and its roots in the LVOV-Warsaw school. Studies in Logic, Grammar and Rhetoric.
[3] Boudry, M., Paglieri, F., & Pigliucci, M. (2015). The Fake, the Flimsy, and the Fallacious: Demarcating Arguments in Real Life. Argumentation.
[4] Nieminen, P., & Mustonen, A.-M. (2014). Argumentation and fallacies in creationist writings against evolutionary theory. Evolution: Education and Outreach
[5] Sahlane, A. (2012). Argumentation and Fallacy in the Justification of the 2003 War on Iraq. Argumentation

Methodology
Categories

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- We studied various fallacy types from the related textbooks [1, 2, 3] and decided on a set of some 20 fallacy types as our classes.
- Prepared a guideline

ad hominem	false analogy	slippery slope	appeal to authority
hasty generalization	false cause	sunk costs	straw man
biased sample	begging the question	appeal to force	red herring
anecdotal argumentation	appeal to ignorance	appeal to popularity	appeal to sarcasm
composition /division	appeal to consequences	appeal to emotions	trivializing

[1] Damer, T. E. (2008). Attacking Faulty Reasoning: A Practical Guide to Fallacy-Free Arguments. Wadsworth Cengage Learning, Belmont, CA.
[2] Govier, T. (2010). A Practical Study of Argumentation. Wadsworth Cengage Learning, Belmont, CA.
[3] Tindale, C. W. (2007). Fallacies and Argument Appraisal. Cambridge Univ. Press, New York, NY.

Methodology
Categories

- Our unit of analysis is sentences in documents, assuming each sentence is a potential argument.

An example text (written by a user who is against homeschooling):

S0 Well this has been fun but I have to go back to work.
S1 **Remember, keeping your kids away from knowledge that you don’t like is a moral crime.**
S2 Have a nice day.

Methodology
Data

- Determining the domain, type and source of data
- Building the ground truth
- Increase the amount of labelled data

Preliminary Annotations I

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- 15 texts from the argument-component annotated set of blogpost comments (from the corpus created by [1])
- 86 sentences / 2 expert annotators / 20 types of fallacies
- Multiple annotations per item allowed

Agreement results based on Krippendorff's alpha:

MASI set distance metric	-0.0233
set distance metric	-0.0234
nominal distance metric	-0.0309

[1] Habernal, I. and Gurevych, I. (2015). Argumentation mining in user-generated web discourse. Computational Linguistics. *Under review*.

Preliminary Annotations I

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- The annotations are not reliable, there is a systematic disagreement

- The problems we could detect:
    - The texts are informal and this informal tone is sometimes confused for logical flaws or linguistic ambiguities.
    - Annotators' approach: the level of being picky and detailed, lack of training
    - The number of categories, which is 20, is reasonably high to choose among and remember.

Preliminary Annotations II

- 20 texts from the same collection
- 126 sentences / the same 2 expert annotators / **16** types of fallacies
    - Removed the types: hasty generalization, anecdotal argumentation, slippery slope, trivializing
- Multiple annotations per item allowed
- Took relatively short time

Agreement results based on Krippendorff's alpha:

MASI set distance metric	0.1967
set distance metric	0.1969
nominal distance metric	0.1969

Preliminary Annotations III

- 50 texts from nytimes comments [1] about 7 different topics
- More than 200 sentences / the same 2 expert annotators / previous **16** types of fallacies
- Multiple annotations per item allowed
- Used an annotation tool, webanno [2]

- We decided not to use this source as the texts are relatively well-written / sound

[1] Room for Debate - NYTimes.com, http://www.nytimes.com/roomfordebate
[2] Yimam, S.M., Gurevych, I., Eckart de Castilho, R., and Biemann C. (2013): WebAnno: A Flexible, Web-based and Visually Supported System for Distributed Annotations. In Proceedings of ACL-2013.

Building the Gold Data

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- User comments / discussion pieces from procon [1], an online debate portal

An example text (written by a user to support vegetarianism):

"It’s healthy and it doesn't harm the earth. I don't believe in violence so eating animals tends to harm them, everything deserves to live."

[1] Procon.org - Pros and Cons of Controversial Issues, http://www.procon.org/

Gold Data

- 159 documents
  - Combined two sets of annotations: 1) 59 texts 2) 103 texts

- 2 annotators / 654 sentences / (1)16 (2)15 types of fallacies

ad hominem	appeal to consequences	appeal to emotion	appeal to false authority
appeal to force	appeal to ignorance	appeal to popularity	appeal to sarcasm
begging the question	biased sample	composition / division*	false cause
false cause	red herring	straw man	sunk costs

Gold Data

- 159 documents
  - Combined two sets of annotations: 1) 59 texts 2) 103 texts

- 2 annotators / 654 sentences / 16 types of fallacies

Agreement values

Set	# fallacious sentences / # total sentences	# labels used / # total labels	Krippendorff's alpha	Cohen's kappa
59 texts	46 / 266	11 / 16	0.0842	0.0899
103 texts	47 / 539	13 / 15	0.151	0.1528

Increasing the Amount of Labelled Data

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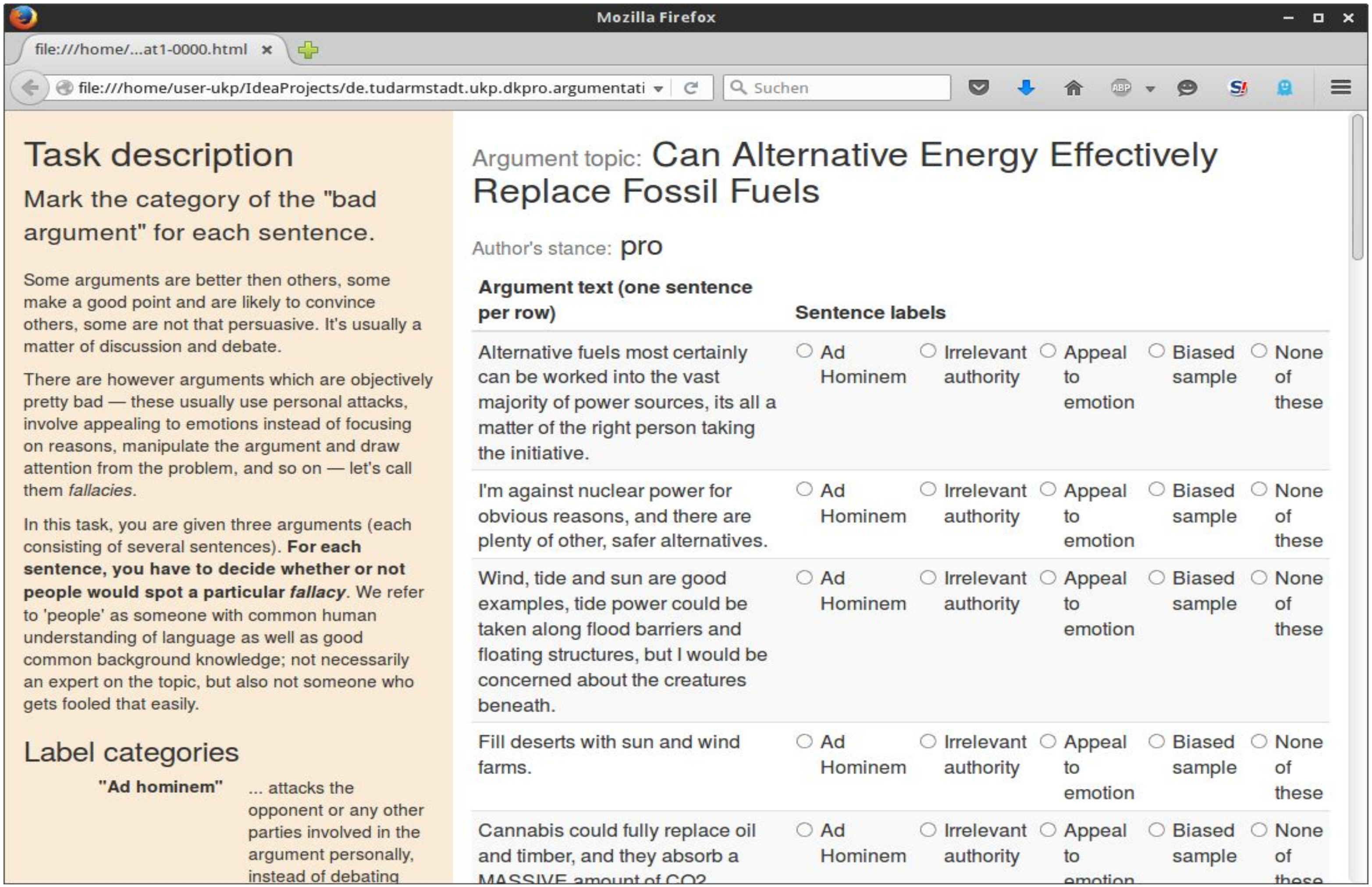
- Either in a long time with experts or in a short time with non-expert people
- We opted for crowdsourcing on mechanical turk [1]

- To have a quality assurance, deployed the texts in our gold data, had them annotated in around 1 week
    - 159 docs / 805 sentences
    - 4 + 4 types of fallacies:
        - Cat1: Ad Hominem, Irrelevant authority, Appeal to emotion, Biased sample
        - Cat2: Appeal to consequences, Appeal to popularity, Appeal to sarcasm, Straw man
    - 106 hits / 25 annotators

[1] Amazon Mechanical Turk, https://www.mturk.com/mturk/welcome

Using Mturk


Mozilla Firefox
file:///home/...at1-0000.html
file:///home/user-ukp/IdeaProjects/de.tudarmstadt.ukp.dkpro.argumentati
Suchen
Task description
Mark the category of the "bad argument" for each sentence.
Some arguments are better then others, some make a good point and are likely to convince others, some are not that persuasive. It's usually a matter of discussion and debate.
There are however arguments which are objectively pretty bad — these usually use personal attacks, involve appealing to emotions instead of focusing on reasons, manipulate the argument and draw attention from the problem, and so on — let's call them fallacies.
In this task, you are given three arguments (each consisting of several sentences). For each sentence, you have to decide whether or not people would spot a particular fallacy. We refer to 'people' as someone with common human understanding of language as well as good common background knowledge; not necessarily an expert on the topic, but also not someone who gets fooled that easily.
Label categories
"Ad hominem"
... attacks the opponent or any other parties involved in the argument personally, instead of debating
Argument topic: Can Alternative Energy Effectively Replace Fossil Fuels
Author's stance: pro
Argument text (one sentence per row)
Sentence labels
Alternative fuels most certainly can be worked into the vast majority of power sources, its all a matter of the right person taking the initiative.
I'm against nuclear power for obvious reasons, and there are plenty of other, safer alternatives.
Wind, tide and sun are good examples, tide power could be taken along flood barriers and floating structures, but I would be concerned about the creatures beneath.
Fill deserts with sun and wind farms.
Cannabis could fully replace oil and timber, and they absorb a
Ad Hominem
Irrelevant authority
Appeal to emotion
Biased sample
None of these

Mturk Annotations I

- Accuracy of majority-voted mturk annotations relative to the gold labels:
  - Cat1: 0.861
  - Cat2: 0.867

- Krippendorff's alpha measure per hit:
  - 106 hits each involving 3 annotators (same people per hit)
  - The agreement values range from -0.2914 to 0.8323..

- Annotators' performance
  - Disagreement rates for each annotator calculated
  - 4 spammers detected
  - A probabilistic quality model [1] applied; annotator quality values range from 0 to 0.5687.

[1] Dirk Hovy, Taylor Berg-Kirkpatrick, Ashish Vaswani, and Eduard Hovy (2013): Learning Whom to Trust with MACE. In: Proceedings of NAACL-HLT 2013.

Mturk Annotations I

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- Accuracy of mturk annotations after removing spammer answers

	nitems		Krippendorff's alpha		Cohen's kappa	
Annotations	category1	category2	category1	category2	category1	category2
full set	805	805	0.0948	0.0962	0.1189	0.1117
sentences without spammer answers	169	154	0.0551	0.0688	0.1122	0.1004
only sentences containing spammer answers	636	651	0.1063	0.1007	0.1223	0.1131

Lessons learned

- Annotation of fallacies by non-experts is not impossible
- The rate of possibly fallacious sentences in the dataset should be as high as possible

Obtaining the Texts Likely-To-Be-More-Fallacious

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- Distant supervision is a method, like bootstrapping, for enlarging the set of labelled data using some heuristic with little learning or manual intervention, first proposed by [1]
- We apply a simple heuristic to filter out non-fallacious documents
- Sources of texts:
  - debate portals:
    - convinceme.net, createdebate.com
    - 24325 discussions threads, a total of 411521 comments
  - disqus:
    - disqus.com
    - 142654 discussion threads, more than some 3M comments

[1] Mintz, M., Bills, S., Snow, R., and Jurafsky, D. (2009). Distant supervision for relation extraction without labeled data. In Proceedings of the AFNLP - ACL ’09, Stroudsburg, PA, USA.

Algorithm for Distant Supervision

Algorithm 1 Distant supervision algorithm for selecting candidate fallacious comments

```
Collect comments per fallacy type where the comments are supposed to be committing that fallacy type if its child
comments contains a keyword for the fallacy.
Require: commentList: list of comments retrieved from an online source; keywords: list of fallacy keywords.
for each keyword in keywords do
    keyword.candidateCommentList ← []                                  ▷ initialize a list per keyword
end for
for each keyword in keywords do
    for each comment in commentList do
        if keyword.string occurs in comment.text then
            parentComment ← comment.parent
            if parentComment is not null then
                if parentComment.parent is null then       ▷ discard the candidates which are deep in the discussions
                    keyword.candidateCommentList.append(parentComment)
                end if
            end if
        else
            keyword.singleReferrentCommentList.append(comment)        ▷ keep this list for further use, maybe for
qualitative analysis or error analysis. not used for annotations
        end if
    end for
end for
```

Results of Distant Supervision

source	# of comments extracted to be fallacious	# all comments
debate portals	632	411521
disqus		

- We held a small annotation study and annotated 20 comments for each fallacy type from a predefined set
  - The comments were fallacious
  - It is hard to capture the context for some of the texts

Mturk Annotations II

- The aim
  - filter out non fallacious text
  - have the fallacious ones re-annotated in another crowdsourcing task
- Dataset: comment set provided by [1]
  - 289 documents, 2398 sentences
- Fallacy types: ad hominem, appeal to consequences, appeal to emotion, appeal to sarcasm and hasty generalization (this comprises biased sample and appeal to popularity)
- Sentence-level annotation
- Mturk
  - 289 hits created, 31 mturkers took part
  - Took around 3 weeks to complete

[1] Habernal, I. and Gurevych, I. (2015). Argumentation mining in user-generated web discourse. Computational Linguistics. *Under review*.

Mturk Annotations II

- Agreement per hit calculated as before
  - Ranges from -0.45 to 0.6.
  - Generally lower than those of our previous task

- The probabilistic quality estimation model [1] applied
  - 929 sentences were predicted to be fallacious (39%)
  - 211 documents with a total of 1906 sentences can be annotated again such that each of these documents contain at least one fallacious sentence.
  - Thus, next annotation study is expected to have higher agreement rates and result in more reliable annotations.

[1] Dirk Hovy, Taylor Berg-Kirkpatrick, Ashish Vaswani, and Eduard Hovy (2013): Learning Whom to Trust with MACE. In: Proceedings of NAACL-HLT 2013.

Summary

- Fallacy detection is a difficult task
- Annotation of fallacies requires a very good understanding of fallacies
  - Determining the set of fallacies
  - Possibly non-overlapping categories with target examples
    - Example case: false cause and hasty generalization
  - Annotator training

- Crowdsourcing
  - Applicable
  - A well-defined task and a concise guideline is important
  - Ways not to waste spammed annotations should be devised
  - The instances to be deployed should be as balanced as possible in terms of the target labels